tomo

AI COACHING PLATFORM

Platform Overview for Athletes

A complete guide to every feature, screen, and intelligent system in Tomo -- what you see, what you can do, and how it all works together to help you train smarter, recover better, and grow as an athlete.

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1 What is Tomo?

Tomo is an AI-powered coaching platform built for young athletes (ages 13-25). It combines daily wellness tracking, intelligent scheduling, personalized training programs, and a conversational AI coach into a single app that adapts to your life -- balancing sport, school, recovery, and personal growth.

At its core, Tomo does three things:

UNDERSTANDS YOU

Every check-in, test, session, and wearable reading builds a real-time picture of your readiness, load, and development trajectory.

ADVISES YOU

AI-powered recommendations tell you what to train, when to rest, and how to balance academics -- personalized to your sport, position, age, and current state.

PROTECTS YOU

9 safety guardrails automatically cap training load, block risky exercises during growth spurts, and flag injury risk before problems happen.

GROWS WITH YOU

Longitudinal tracking of speed, strength, endurance, agility, and mastery pillars builds your athlete identity and professional CV over time.

2 App Structure at a Glance

The app has three main tabs, plus a set of supporting screens accessible from headers and navigation links.

TAB	NAME	PURPOSE
Left	Plan (Timeline)	Your daily schedule -- training blocks, study sessions, exams, recovery. Navigate day-by-day.
Centre	Tomo Chat	AI coaching assistant. Ask anything, log tests, build sessions, manage your calendar. Raised button with Tomo logo.
Right	Dashboard (Trend)	Your command centre. Readiness signal, mode selector, recommendations, performance metrics, weekly trends.

Supporting Screens

SCREEN	ACCESS POINT	WHAT IT DOES
Check-In	Header button (red dot if overdue)	7-step daily wellness wizard
Profile	Header avatar	Level, points, streak, settings, wearables
My Rules	Dashboard or Profile	Configure school hours, sleep, training categories, athlete mode
Notifications	Header bell icon	Priority-sorted alerts by category
Add Event	FAB button on Plan tab	Quick event creation (training, match, study, exam)
Drill Detail	Chat or Programs	Execute a drill with timer, sets, and rest periods
Player CV	Chat or Profile	Build and export your professional athlete CV
PHV Calculator	Chat or Profile	Calculate your growth maturity stage

3 Your Daily Journey

Here is what a typical day looks like for a Tomo athlete, from waking up to winding down.

3.1 Morning: Open the App

When you open Tomo, the app loads your entire state in one shot. Behind the scenes, 18 parallel queries run to fetch your profile, snapshot, schedule, check-in history, recommendations, benchmarks, wearable data, and more. This takes under 2 seconds.
You land on the **Dashboard** -- your daily command centre. At the top, you see your **Athlete Mode** (Balanced, League, Study, or Rest) and the readiness signal derived from yesterday's data.

> **If you haven't checked in yet**
> The dashboard shows a BASELINE signal (grey) and the check-in button in the header shows a red dot. Your first action should always be to check in.

3.2 Check In

The daily check-in is a 7-step wellness wizard that takes about 30 seconds:

STEP	WHAT YOU RATE	HOW
1	Sleep quality	Emoji scale (1-10)
2	Energy level	Emoji scale (1-10)
3	Stress level	Emoji scale (1-10)
4	Soreness	Emoji scale (1-10)
5	Pain flag	Yes / No (safety trigger)
6	Mood	Emoji scale (1-10)
7	Motivation	Emoji scale (1-10)

Tapping an emoji auto-advances to the next step. On completion, a confetti animation plays and your **readiness score** is computed instantly:

GREEN -- GOOD TO GO
Full training is safe. All systems are recovered. Train at full intensity.

YELLOW -- PROCEED WITH CARE
Some stress markers are elevated. Training is fine but high-intensity work should be moderated.

RED -- RECOVERY PRIORITY

Multiple markers are suppressed. High-intensity training is blocked. Focus on mobility, light movement, and recovery.

> **Pain Flag Safety Gate**
>
> If you report pain at step 5, Tomo immediately triggers a REST recommendation and flags injury risk. Your readiness drops to RED regardless of other scores, and high-intensity programs are blocked until the next clean check-in.

3.3 Dashboard -- Your Command Centre

After check-in, the Dashboard updates in real-time. It is built from CMS-managed sections that your coach or performance director can reorder, enable, or disable. The standard layout has four zones:

ZONE	SECTION	WHAT YOU SEE
1	Athlete Mode Hero	Current mode pill (Balanced/League/Study/Rest), quick-switch pills, signal badge
2	Daily Recommendations	Priority-sorted recommendation cards (expandable): readiness, load warnings, recovery, development, academic, motivation
3	CMS Sections	Welcome card, protocol alerts, growth tracking, readiness ring, core vitals (KPI row), weekly trend sparklines, dual load index, benchmark progress, AI recommendations, upcoming events, consistency streak bar
4	Up Next	Your next scheduled activities with contextual AI hints

The Readiness Ring

A circular animated ring shows your readiness score (0-100). Green ring for scores above 70, yellow for 40-70, red below 40. Below the ring, a trend indicator shows how today compares to yesterday (e.g. "+8 vs yesterday").

Core Vitals Row

Compact KPI tiles showing your key numbers at a glance: sleep hours, HRV, energy, soreness, ACWR, and dual load index. Each tile includes a sparkline showing the 7-day trend.

3.4 Plan Your Day

Switch to the **Plan** tab to see your full daily timeline. Events are laid out vertically by time:

- **School blocks** -- locked, shown in grey, cannot be moved

- **Training sessions** -- club, gym, or personal, with intensity badges
- **Study blocks** -- scheduled study time with subject tags
- **Matches** -- fixture events with pre/post buffers
- **Recovery** -- auto-generated after matches or when readiness is RED

Tap any event to edit it. Tap a gap in the timeline to add a new event. Use the floating action button (+) to quickly create training, study, or other blocks.

> **Smart Conflict Detection**
>
> If you try to schedule a HARD session the day before a match, or squeeze two sessions into a time slot that violates your buffer rules, Tomo flags the conflict and suggests alternative times.

3.5 Train

When it is time to train, Tomo can build you a personalized session. Ask Tomo Chat "build me a session" or tap the training block on your timeline. The AI considers your readiness, position, benchmark gaps, and current load to select drills and set appropriate volumes.
During a drill, the **Drill Detail** screen provides a guided timer: countdown to start, active phase (counting down each set), rest phases between sets, and a completion screen where you log touches, reaction times, and RPE.

3.6 Study

Tomo treats academics as seriously as training. The exam planner lets you input exam dates and subjects, and Tomo generates a study schedule that respects your training load. In **Study Mode**, study blocks are extended by 50% and training intensity is capped on exam days.

3.7 Reflect & Wind Down

After sessions, Tomo may nudge you to log feedback (workout quality, fatigue level, notes). Journal entries build your coaching profile and contribute to your CV narrative. The notification engine tracks whether you have reflected and gently prompts you if you have not.

4 The Three Tabs in Detail

4.1 Plan (Timeline Tab)

The Timeline is your single-day planning view. Key features:

- **Day navigation** -- left/right arrows to move between days
- **Readiness badge** -- Green/Yellow/Red indicator at top
- **Balance bar** -- visual breakdown of sleep, stress, and fatigue
- **AI coaching card** -- context-aware tip for the day
- **Lock-In card** -- daily commitment pledge for accountability
- **Event blocks** -- colour-coded by type (training, study, exam, recovery, match)
- **Exam planner** -- visible when exam period is active in your rules

4.2 Tomo Chat (AI Coach)

The centre tab is your AI coaching assistant. Tomo understands 60+ intents across three specialised agents:

TIMELINE AGENT

Calendar operations: "What's my schedule this week?", "Add gym tomorrow at 5pm", "Move my study block to Thursday", "Any conflicts this week?"

OUTPUT AGENT

Performance and health: "What's my readiness?", "Log a sprint test", "Build me a technical session", "Show my strengths and gaps", "Calculate my PHV"

MASTERY AGENT

Development and identity: "Show my achievements", "How consistent have I been?", "Update my CV", "What's my test trajectory?"

Chat features include:

- **Suggestion chips** -- tap to send pre-built messages
- **Voice input** -- speak your message instead of typing
- **Streaming responses** -- real-time typewriter effect
- **Confirmation cards** -- for write actions (create event, log test), Tomo shows a preview card and asks you to confirm before saving
- **Rich response cards** -- stat grids, schedule views, benchmark comparisons, program recommendations
- **Session continuity** -- Tomo remembers your conversation context within a session

4.3 Dashboard (Trend Tab)

The Dashboard is your at-a-glance command centre. It renders up to 14 section types, managed by your performance director via CMS:

SECTION	TYPE	WHAT IT SHOWS
Athlete Mode Hero	signal_hero	Active mode, quick-switch pills, signal urgency badge
Daily Recommendations	daily_recs	Top 6 priority-sorted recommendation cards
Welcome Card	custom_card	Personalised greeting with name interpolation
Protocol Banner	protocol_banner	Active safety protocols (mandatory/advisory/info)
Growth Tracking	growth_card	Development narrative from AI
Readiness Ring	status_ring	Animated score ring with yesterday comparison
Core Vitals	kpi_row	Compact KPI tiles (sleep, HRV, energy, soreness, ACWR)
Weekly Trends	sparkline_row	7-day sparkline charts for key metrics
Dual Load Index	dual_load	Combined academic + athletic load gauge (0-100)
Benchmark Progress	benchmark	Overall percentile and position-specific test gaps
AI Recommendations	rec_list	Expandable AI recommendation cards
Upcoming Events	event_list	Next scheduled events from calendar
Consistency Streak	engagement_bar	Visual streak progress bar
Up Next Timeline	up_next	Today's remaining schedule with AI context

5 Athlete Modes

Athlete Mode is a single setting that cascades through every system in Tomo. When you switch modes, your schedule rules, training intensity caps, program recommendations, AI coaching tone, and dashboard priorities all adjust automatically.

MODE	WHEN TO USE	WHAT CHANGES
Balanced	Normal weeks with no special pressures	Default settings. Full training intensity. Standard study/training balance. Coaching tone: balanced.
League Season	During competitive fixture weeks	Recovery days auto-added after matches. Max HARD sessions reduced. Post-match buffers enforced. Gym days may reduce. Coaching tone: performance-focused.
Study Focus	During exam preparation periods	Study blocks extended by 50%. Training intensity capped on exam days. Academic recommendations boosted to top priority. Personal development sessions dropped. Coaching tone: academic/supportive.
Rest & Recovery	After illness, injury, or planned deload	Load cap at 50%. High-intensity categories blocked (plyometrics, speed, explosive). Recovery and mobility prioritised. Coaching tone: supportive.

What Each Mode Controls (12 Parameters)

PARAMETER	BALANCED	LEAGUE	STUDY	REST
Max HARD sessions/week	3	2	2	1
Max sessions/day	2	2	2	1
Study duration multiplier	1.0x	1.0x	1.5x	1.0x
Gym days reduction	None	Cap at 3	Cap at 2	Cap at 1
Drop personal dev	No	No	Yes	No
Exam day intensity cap	None	None	LIGHT	REST
Recovery after match	No	Yes	No	Yes
Load cap multiplier	1.0	0.85	0.7	0.5
AI coaching tone	Balanced	Performance	Academic	Supportive
Priority boosts	None	+Recovery	+Academic	+Recovery

Where Mode is Applied (All 8 Systems)

1. **Schedule Rule Engine** -- adjusts gym days, study duration, personal dev, intensity caps
2. **Schedule Validation** -- enforces mode rules when creating/editing events
3. **AI Orchestrator** -- injects mode-aware rules and coaching tone into the AI system prompt
4. **Program Guardrails** -- applies mode load cap and blocks categories (e.g. rest blocks plyometrics)
5. **Deep Program Refresh** -- AI program selection respects mode load cap and coaching tone
6. **Boot Recommendations** -- mode priority boosts re-sort recommendation ordering
7. **Load Warning Computer** -- rest mode lowers warning thresholds
8. **Recovery Computer** -- league mode forces post-match recovery

6 Intelligent Systems Behind the Scenes

You do not see these systems directly, but they power every recommendation, every safety guard, and every personalized response you receive.

6.1 Readiness Engine

Your readiness is computed from your daily check-in, wearable data (if connected), and historical trends. The engine produces a score (0-100) and a RAG status (Red/Amber/Green) that determines what training is safe today.

```
Check-in (7 inputs) + Wearable data (HRV, sleep, HR)
  → Readiness Score (0-100)
  → RAG Status: GREEN (≥70) | YELLOW (40-69) | RED (<40)
  → Intensity Recommendation: HARD | MODERATE | LIGHT | REST
```

6.2 Schedule Rule Engine

A pure rules engine that governs all scheduling decisions. It uses a strict priority order:

PRIORITY	CATEGORY	NOTES
P1	School	Non-negotiable, locked blocks
P1	Exams	Non-negotiable, locked blocks
P2	Match	League/fixture, time set by club
P3	Recovery	Mandatory post-match or RED readiness
P4	Club Training	Coach-set times
P5	Gym	Individual strength & conditioning
P6	Study	Daily academic block
P7	Personal Dev	Film study, tactical review, mental skills

Hard constraints enforced at all times:

- Maximum 3 HARD sessions per week (adjustable by mode)
- Maximum 2 sessions per day
- No HARD training the day before a match
- No HARD training on exam days
- Mandatory recovery day after matches
- 30-minute default buffer between events; 45 min after high intensity; 120 min before a match

6.3 Recommendation Intelligence Engine (RIE)

Nine specialized "computers" generate personalized recommendations:

COMPUTER	WHAT IT PRODUCES
Readiness	Energy/soreness/sleep analysis, recovery urgency
Recovery	Sleep debt detection, HRV suppression alerts, soreness trend warnings
Load Warning	ACWR monitoring, dual load detection, overtraining risk
Academic	Exam proximity alerts, study planning guidance
Motivation	Streak maintenance nudges, consistency tracking
Development	Benchmark gap analysis, targeted program suggestions
Triangle Alert	Combined readiness + load + wellness risk flags
CV Opportunity	Performance highlights, personal records, recruitment narrative
Journal Nudge	Pre/post-session reflection prompts

Recommendations are priority-sorted (P1 = critical, P4 = informational) and filtered by your athlete mode. In Study mode, academic recommendations are boosted to the top; in League mode, recovery recommendations take priority.

6.4 Program Guardrails (9 Safety Rules)

Every training program you see has passed through 9 deterministic safety checks:

#	RULE	WHAT IT DOES
1	ACWR Load Gate	If your training load ratio is dangerously high (>1.5), caps volume at 60%
2	Readiness Gate	RED readiness caps load at 50% and blocks explosive/HARD categories
3	HRV Suppression	If HRV is <85% of your baseline, reduces load proportionally
4	Sleep Quality	Poor sleep (<24h ago) reduces training volume by 20%
5	Dual Load	High combined academic + athletic load caps training at 70%
6	Injury Risk	High injury risk: 60% load cap + injury prevention becomes mandatory
7	Wellness Trend	Declining wellness over 7 days triggers volume reduction
8	Training Age	Beginners (<12 weeks) get 60% load cap and blocked from advanced exercises
9	Athlete Mode	Mode-specific load cap (rest mode = 50%) and category blocking

> **These rules cannot be overridden**
>
> Guardrails are applied after program selection (whether by AI or by a coach). Even if the AI recommends a high-intensity program, the guardrails will automatically reduce volumes or block it entirely if your current state makes it unsafe.

6.5 Protocol Layer (PDIL)

The Performance Director Intelligence Layer evaluates CMS-authored "protocols" against your current state. Protocols are condition-based rules that your coach or performance director sets up -- for example:

- "If ACWR > 1.3 AND readiness is RED, enforce REST-only training and notify athlete"
- "If PHV stage is mid-PHV, block barbell squats and Olympic lifts"
- "If exam_proximity_score > 80, cap intensity at LIGHT"

Protocols can be mandatory (cannot train at all until resolved), advisory (training allowed with modifications), or informational. Safety-critical protocols force the AI to use a more careful model for responses.

6.6 Notification Engine

Notifications are generated automatically by event handlers and delivered to your notification centre. Key features:

- **7 categories**: Critical, Readiness, Recovery, Load Management, Achievements, Recommendations, Journal
- **Deduplication**: Same alert type is updated, not duplicated (no "10 readiness RED" spam)

- **Fatigue guard**: If you dismiss a notification type 3 times in 24 hours, Tomo stops sending it
- **Auto-expiry**: Notifications resolve when the underlying condition changes (ACWR drops, readiness improves)
- **Actionable**: Each notification has a primary action (e.g. "Log check-in", "See program", "Review recovery plan")

7 AI Chat -- How Tomo Understands You

Tomo Chat is not a generic chatbot. It is a three-agent system grounded in your real-time data, schedule rules, and performance context.

How a Message is Processed

```
You type a message
  → Layer 1: Exact match (~170 phrases, instant, $0)
  → Layer 2: Haiku AI classification (~60 intents, ~200ms)
  → Layer 3: Full orchestrator with tools (complex queries)
  → Route to Timeline / Output / Mastery agent
  → Agent calls tools (get_readiness, create_event, log_test...)
  → Format response (headline + stat cards, no wall of text)
  → Stream back to you
```

What Tomo Knows When Responding

Every time Tomo responds, the AI has access to:

- Your sport, position, age band, and gender
- Current readiness score and RAG status
- Today's schedule and upcoming events (7 days)
- ACWR and load metrics
- Benchmark percentiles, strengths, and gaps
- Active recommendations from the RIE
- Your schedule rules and available hours
- Your athlete mode and its parameters
- Active PDIL protocols and safety flags
- Wearable data (HRV, sleep, heart rate)
- Conversation history from the current session

What You Can Ask

CATEGORY	EXAMPLE MESSAGES
Readiness	"What's my readiness?", "How am I doing?", "Am I ready to train?"
Schedule	"What's today?", "Plan my week", "Add gym Thursday at 5pm"
Training	"Build me a session", "Show me drills for speed", "What should I train?"
Testing	"Log a sprint test", "Record my jump", "What's my 30m time?"
Performance	"Show my strengths and gaps", "Where do I rank?", "What's my ACWR?"
Programs	"Show my programs", "What programs are recommended?"
Academic	"Plan my study week", "I have exams in 2 weeks"
Recovery	"I'm feeling tired", "Legs are dead", "Should I rest today?"
CV/Identity	"Update my CV", "Show my achievements", "What's my consistency?"
Growth	"Calculate my PHV", "Am I in a growth spurt?"

Safety in Chat

- **Write actions require confirmation**: Creating events, logging check-ins, and saving tests show a preview card. You must confirm before anything is saved.
- **RED readiness blocks intensity**: If you ask to build a HARD session while RED, Tomo will refuse and suggest recovery activities instead.
- **PHV safety**: For athletes in mid-PHV (growth spurt), Tomo appends safety warnings about high-impact exercises.
- **Gen Z tone**: Responses are calibrated to your age band. U13 gets simpler language, U17 gets direct coaching, U19+ gets professional performance advice.

8 Testing & Performance Tracking

Tomo offers both phone-based tests (using your device's sensors) and manual test entry. All results feed into your benchmark profile and influence program recommendations.

Phone-Based Tests

TEST	WHAT IT MEASURES	HOW
Sprint	10m, 20m, 30m, flying 20m times	Accelerometer-based motion detection
Jump (CMJ)	Counter-movement jump height	Accelerometer flight time calculation
Agility	505, T-test, 5-10-5, Illinois, Arrowhead	On-screen cues + motion detection
Balance	Single-leg stability time	Gyroscope-based wobble detection
Reaction	Visual/auditory reaction time	On-screen stimulus + tap timing

Manual Test Entry

You can log results from any test via Tomo Chat ("log my sprint test") or via the fitness tests screen in your profile. Each result is automatically compared against normative data for your age band and position.

Benchmark Profile

Your test results build a benchmark profile showing:

- **Overall percentile** -- how you compare to peers at your age and position
- **Strengths** -- test categories where you rank above average (highlighted green)
- **Gaps** -- categories where you need development (highlighted for targeted programs)
- **Trajectory** -- improvement arc over time for each test type

The benchmark profile directly drives the AI program selection -- weak areas receive higher-priority programs.

9 Programs & Training

Tomo's training program system combines a 31-program catalog with AI-powered personalisation and deterministic safety guardrails.

How Programs Are Selected

> 1. AI analyses your full context (tests, vitals, PHV, readiness, load, position)
> 2. Selects 8-15 best-fit programs from 31-program catalogue
> 3. Assigns priority: mandatory (3-5), high (3-5), medium (2-5)
> 4. Personalises impact statements using YOUR data
> 5. 9 guardrail rules applied -- volumes adjusted, unsafe programs blocked
> 6. Final list cached for 24 hours, shown in Programs panel

Program Categories

- Speed and acceleration
- Strength and conditioning
- Agility and change of direction
- Plyometric and explosive power
- Endurance and aerobic base
- Injury prevention (Nordic, ACL, ankle, groin, hip)
- Mobility and flexibility
- Position-specific drills
- Match preparation
- Recovery and deload

Position Matrix

Each position has a training matrix defining mandatory programs (must do), recommended programs (should do), and weekly structure (how many sessions of each type per week). For example, a football winger has mandatory speed/acceleration work but lower strength emphasis than a centre-back.

Coach-Assigned Programmes

Your coach can publish programmes targeted at you specifically, your position group, or all athletes. These appear alongside AI-recommended programs but are marked as coach-assigned. The same 9 guardrails still apply.

10 Wearable Integration

Tomo integrates with wearable devices to enrich your data picture with objective measurements.

WHOOP

- **Connection**: OAuth flow from Settings screen. One-tap authorise.
- **Data synced**: Sleep hours, sleep quality, HRV (morning), resting heart rate, recovery score, strain score
- **Priority**: When WHOOP data is available, it takes priority over manual check-in values for sleep. Displayed as "Source: Whoop" in your vitals.
- **Sync**: Automatic on app launch. Manual sync button available in Settings.
- **Data transparency**: Full Whoop data view accessible from Settings to see exactly what was synced.

Apple Watch (HealthKit)

- **Connection**: HealthKit permission from Settings screen.
- **Data synced**: Steps, heart rate, workouts (type, duration, calories), sleep analysis

Wearable data feeds directly into the readiness engine, HRV suppression guardrail, and sleep quality assessments. If your WHOOP shows HRV at 65% of baseline, the program guardrails automatically reduce your training load without you needing to do anything.

11 Player CV & Identity

Your performance data, achievements, and reflections build a professional athlete CV that you can export and share.

CV Sections

SECTION	CONTENT	SOURCE
Personal Statement	AI-generated or manually written narrative about who you are as an athlete	AI draft + manual edit
Career History	Club entries with dates, appearances, goals	Manual entry
Education	Academic qualifications and institution	Manual entry
Performance Data	Benchmark percentiles, test scores, training age	Auto from tests
Media	Video links (highlights, full match, training footage)	Manual upload
References	Coach/academy director testimonials	Manual entry
Character Traits	Leadership, languages, awards, camp attendance	Manual entry

Two CV variants are available: **Club** (for trials and scouting) and **University** (for academic sporting scholarships). Both can be exported to PDF or shared via a link.

12 Gamification & Streaks

Tomo uses a points, levels, and streak system to keep you consistently engaged without making it feel like a chore.

Points & Levels

You earn points for completing check-ins, logging sessions, running tests, and engaging with recommendations. Points accumulate toward levels, and each level unlocks new features or visual milestones displayed on your profile.

Streak System

TIER	DAYS	STATUS
Started	1-3	Just getting going
Building	4-7	Building the habit
Consistent	8-14	Consistently showing up
Dedicated	15-30	Real commitment
Veteran	31-60	Part of your routine
Legend	60+	Elite consistency

Your streak is displayed on the Dashboard (engagement bar section) and on your Profile. Freeze tokens allow you to protect your streak on rest days or when travelling.

13 How Everything Connects

Tomo is not a collection of separate features. Every action you take triggers a chain of updates that ripple across the entire platform. Here is how a single event flows through the system.

Example: You Complete a Training Session

```
1. You log the session (via Chat or Timeline)
   → Event handler processes SESSION_LOG
   → Daily load updated (ACWR recalculated)
   → Session count and training age updated
   → Athlete snapshot (Layer 2) rewritten

2. Notifications fire (async)
   → If ACWR now > 1.3: "Load Warning" notification created
   → Journal nudge: "How did the session go?"

3. Next time you open the app (boot)
   → Fresh snapshot loaded
   → PDIL re-evaluates protocols against new load
   → Guardrails update: future programs adjusted for ACWR
   → Signal evaluates: load warning pill may appear on dashboard
   → Recommendations re-sorted by new priorities

4. You ask Tomo Chat "am I overtraining?"
   → Agent sees updated ACWR in context
   → Responds with load analysis + recovery recommendation
   → Suggests deload if ACWR is in danger zone
```

Example: You Switch to Study Mode

1. You tap "Study" on the mode selector
   → PATCH to schedule preferences
   → Mode change event processed
   → Snapshot updated with new mode
   → Dashboard refreshes instantly (optimistic UI)

2. All downstream systems cascade
   → Schedule rules: study blocks extend 50%, gym days reduce
   → Validation: new events checked against study mode caps
   → AI: coaching tone shifts to "academic", study-focused
   → Guardrails: load cap drops to 70%
   → Programs: AI selects lighter programs, blocks plyometrics
   → Recommendations: ACADEMIC recs boosted to top of list

The Athlete Snapshot -- Single Source of Truth

At the centre of everything is the **athlete snapshot** -- a single database row containing ~80 computed fields that represent your current state. Every event (check-in, session, test, wearable sync) updates the snapshot. Every downstream system (guardrails, recommendations, signals, AI, protocols) reads from it.
This means there is no stale data, no conflicting views, and no system that "doesn't know" about your latest check-in. One truth, everywhere, always current.

The Safety Promise

If anything goes wrong in the system (a database query fails, the AI service is unreachable, a protocol evaluation errors out), Tomo defaults to the most conservative option: lower load caps, stricter intensity limits, and supportive coaching. The system is designed to fail safe, never fail dangerous.

— End of Document —
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